

Brno’s changing surface heat

Summer surface temperature, 2001-2025 | A brief for city decisions

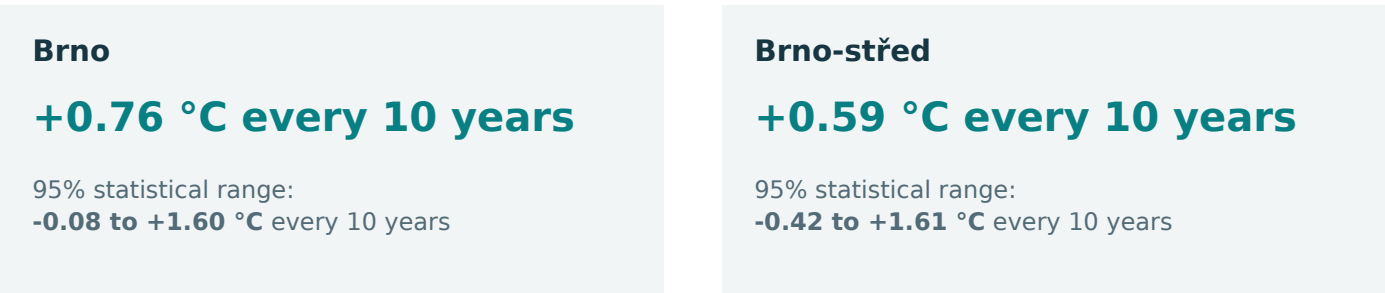
Satellite measurements suggest Brno’s surfaces are warming, but the rate is uncertain. They measure roofs, roads and vegetation; air temperature and effects on people need separate measurements.



01 WHY USE MORNING MEASUREMENTS?

Terra’s morning satellite record spans **2001-2025** and meets our quality checks across **98.5% of Brno and 91.3% of Brno-střed**. Night records leave much larger gaps in the centre; Aqua’s full-summer record starts in 2003. Morning is the most complete comparison here, but does not represent the daily average or afternoon peak. [1]

02 ESTIMATED MORNING CHANGE, 2001-2025



How to read these panels: the large number is the estimated average morning change every 10 years. The range shows statistical uncertainty. Both ranges include zero (no change), so these data do not establish a precise warming rate. The range also changes with the statistical method; this is not proof that no warming occurred. [9]

03 TWO DATA SOURCES, TWO DIFFERENT JOBS

Free MODIS and Landsat archives were processed in **Google Earth Engine** (screening/export) and **Python** (analysis/maps).

MODIS: frequent, coarse measurements. Daily data at about 1 km detail provide the 25-year record. We use June-August, at least 20 clear observations per summer and at least 20 summers per location. Clouds obscure surfaces; suspect pixels are removed, never filled in. [1,5]

Landsat: finer detail, fewer observations. Page 2 compares 2001-2005 with 2021-2025 at the same locations. Landsat 8/9 now offer an 8-day repeat cycle, but cloud-free images are less frequent. More passes do not make thermal imaging all-weather. [2,11]

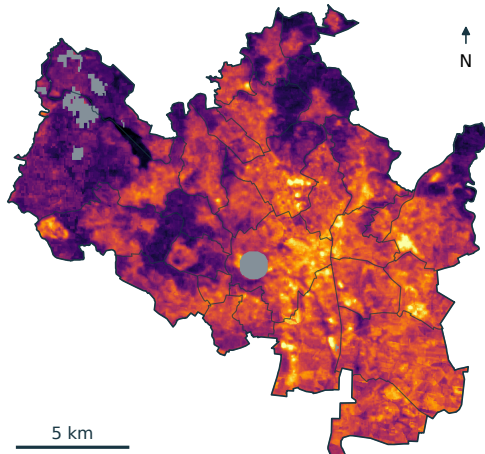
One important limit: Terra’s typical morning observation time shifted from 11:18 to 10:00 over the record. A surface can have different temperatures at different hours, so the estimated trend cannot be attributed entirely to climate change. Night-time results are kept separately in the supporting analysis. [3,4]

Where surface heat changed

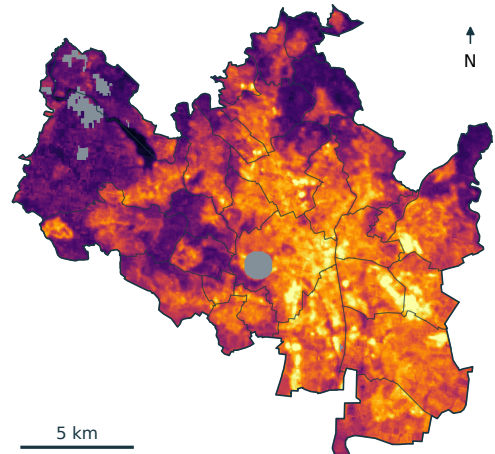
Landsat | 2001-2005 versus 2021-2025 | clear-sky summers

A and B use the same temperature scale and spatial coverage. C subtracts the early temperature from the recent temperature at each location: red is warmer, blue cooler.

A 2001-2005



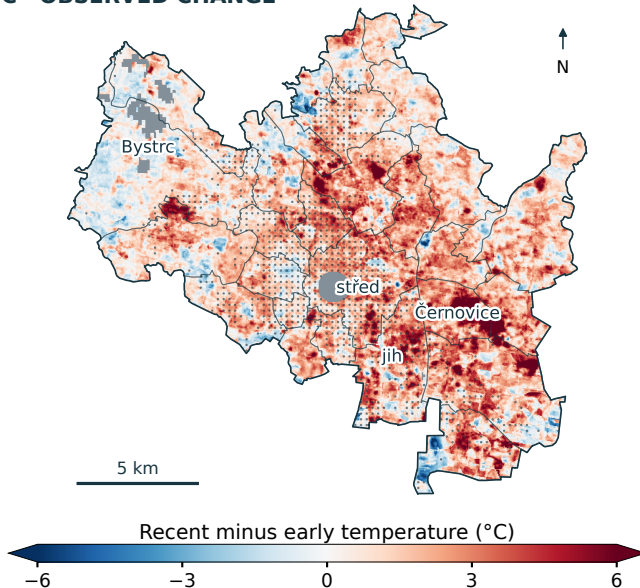
B 2021-2025



Both periods: typical summer surface temperature (°C)

22 26 30 34 38 42

C OBSERVED CHANGE



Recent minus early temperature (°C)

-6 -3 0 3 6

CHANGE AT A TYPICAL LOCATION

+1.39 °C

Middle value of all local changes. Both periods cover 98.2% of Brno.

Brno-střed: +2.19 °C

90.9% of the centre is mapped. A stricter quality check leaves only 19.7%, so this estimate is exploratory.

Local uncertainty. Where both checks apply, changing the quality limit shifts the estimate by typically 1.10 °C. Small changes need caution.

Less strict quality check only: 24.3%

Not enough measurements: 1.8%

White inside C = little change. Colour does not show certainty.

How the maps were made. Each period combines summer measurements from at least 3 years and 5 dates at the same locations; C = B - A. Wider coverage needs a less strict uncertainty limit per measurement (5 °C instead of 3 °C). Weather and sensor differences also affect C. Thermal detail is 60-120 m on a 30 m grid: read city blocks, not individual buildings. [2,7,8,10]

1. Adaptation: reduce heat exposure. Prioritise Brno-střed, Černovice and Brno-jih, where current hot surfaces cluster. Target shade trees at schools, care homes and walking routes; verify benefits with local air measurements. [6]

Next investment. Add ground air/night monitoring. Before buying OroraTech or other surveys, trial calibration, usable cloud-free revisit and cost per usable image in Brno. Satellite maps target checks; they do not measure health or emissions benefits.

2. Mitigation: reduce energy and emissions. Pilot suitable cool/green roofs to reduce cooling demand. Assess annual heating and cooling energy before claiming emissions savings; compare pilot buildings with untreated ones. [6,12]

Sources. [1] MODIS QA · [2] Landsat ST · [3] Terra drift · [4] Aqua drift · [5] Retrieval issues · [6] Cooling measures · [7] Brno districts · [8] OSM / ODbL · [9] Trend uncertainty · [10] Native resolution · [11] Revisit cycle · [12] Roof energy trade-offs